



**MELBOURNE
INSTRUMENTS**

ROTO-CONTROL SYSEX API

Version 1.6

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Revision History

Version	Date	Change
1.0	23/02/2025	First release.
1.1	07/04/2025	Updates to support MACRO params: • LEARN PARAM • PARAM LEARNED • SET MAPPED CONTROL NAME (new)
1.2	23/06/2025	General updates and updates to support Bitwig Studio: • DAW PING RESPONSE • ROTO-DAW CONNECTED (new) • REQUEST ROTO FW VERSION (new) • ROTO FW VERSION (new) • TRACK DETAILS • PLUGIN DETAILS • DAW SELECT PLUGIN • LEARN PARAM • CONTROL MAPPED (renamed, was PARAM LEARNED) • MACRO PLUGIN PAGE INDEX (new) • CONTROL LEARNED (new) • TOGGLE REMOTE PAGE (new) • TOGGLE TRACK GROUPING (new) • DAW SELECT TRACK • REQUEST SEND TRACK NAMES (new) • SEND TRACK NAMES (new) Document formatting improvements.
1.3	05/11/2025	Increased number of tracks to two bytes: • NUM TRACKS • FIRST TRACK • SET FIRST TRACK • TRACK DETAILS • ROTO SELECT TRACK • DAW SELECT TRACK • TOGGLE TRACK GROUPING Added new commands: • REQUEST ROTO SYSEX API VERSION • ROTO SYSEX API VERSION
1.4	26/11/2025	Added Appendix A: Colour Scheme Table
1.5	17/02/2026	Updates to support Logic Pro
1.6	24/07/2026	Added commands to support showing control values and MIX VU meters. Added Appendix B: MIDI CC Mappings

Compatibility

Version	Compatibility
1.0	• ROTO-SETUP v1.0.0 • ROTO-SETUP v1.1.1 • ROTO-SETUP v1.1.2
1.1	• ROTO-SETUP v1.1.3 • ROTO-SETUP v1.1.4
1.2	• ROTO-SETUP v2.0.0
1.3	• ROTO-SETUP v2.1.0
1.4	
1.5	• ROTO-SETUP v2.1.0 • ROTO-SETUP v3.0.1 • ROTO-SETUP v3.1.0 • ROTO-SETUP v3.2.0
1.6	

1. Introduction

The ROTO-CONTROL SYSEX API allows communication and configuration with an external DAW such as Ableton. The interface used is MIDI via USB.

The interface is bi-directional in that commands can be sent to ROTO-CONTROL and received from ROTO-CONTROL asynchronously.

Note 1: For conciseness ROTO-CONTROL is also referred to as ROTO throughout this document.

Note 2: All values are specified in hexadecimal.

1.2 Command Format

All ROTO-CONTROL SYSEX commands take the following format:

F0 00 22 03 02 <data> F7

Where:

- F0 = Start SYSEX message byte.
- 00 22 03 = Melbourne Instruments SYSEX ID.
- 02 = ROTO-CONTROL v1 device ID.
- <data> = The command payload.
- F7 = End SYSEX message byte.

1.3 Commands Overview

Type	Sub-type	Description	To ROTO	From ROTO
0A: GENERAL	01: DAW STARTED	The DAW has started a session with ROTO	Y	N
	02: PING DAW	Ping to check if the DAW is running	N	Y
	03: DAW PING RESPONSE	Ping response from the DAW	Y	N
	04: NUM TRACKS	Number of tracks	Y	N
	05: FIRST TRACK	First track page index	Y	N
	06: SET FIRST TRACK	Set the first track page index	N	Y
	07: TRACK DETAILS	Track details in the current track page	Y	N
	08: TRACK DETAILS END	Finished sending track details	Y	N
	09: ROTO SELECT TRACK	ROTO-Control has selected a track	N	Y
	0A: REQUEST TRANSPORT STATUS	Request the current transport status	N	Y

	0B: TRANSPORT STATUS	The current transport status	Y	N
	0C: ROTO-DAW CONNECTED	The ROTO has received the DAW PING RESPONSE, and the ROTO and DAW are now connected.	N	Y
	0D: REQUEST ROTO FW VERSION	The DAW has requested the ROTO firmware version	Y	N
	0E: ROTO FW VERSION	The ROTO firmware version	N	Y
	0F: REQUEST ROTO SYSEX API VERSION	The DAW has requested the ROTO SYSEX API version	Y	N
	10: ROTO SYSEX API VERSION	The ROTO SYSEX API version	N	Y
	11: SET TRACK DETAILS	Set track details in the current track page and immediately show on ROTO	Y	N
	12: RESET TRACK DETAILS	Remove the track details in the current track page and immediately clear on ROTO	Y	N
	13: SET TRACK COLOUR	Set the track colour in RGB	Y	N
	14: ROTO PAGE LEFT	ROTO page left has been pressed	N	Y
	15: ROTO PAGE RIGHT	ROTO page right has been pressed	N	Y
	16: SET CURRENT TRACK NAME	Set the currently selected track name	Y	N
	17: SET CURRENT TRACK COLOUR	Set the currently selected track name in RGB	Y	N
	18: SHOW CONTROL VALUE	Show the current value of a control on the control screen	Y	N
0B: PLUGIN	01: SET PLUGIN MODE	Set the DAW in PLUGIN mode	N	Y
	02: NUM PLUGINS	Returns the number of PLUGINS on the current track	Y	N
	03: FIRST PLUGIN	First PLUGIN page index	Y	Y
	04: SET FIRST PLUGIN	Set the first PLUGIN page index	N	Y

	05: PLUGIN DETAILS	Details of a PLUGIN	Y	N
	06: PLUGIN DETAILS END	Finished sending PLUGIN details	Y	N
	07: ROTO SELECT PLUGIN	The ROTO-CONTROL has selected a PLUGIN	Y	N
	08: DAW SELECT PLUGIN	The DAW has selected a PLUGIN	Y	N
	09: SET PLUGIN LEARN	Put the DAW in learn mode	Y	N
	0A: LEARN PARAM	A param has been learned	Y	N
	0B: CONTROL MAPPED	Mapped control details	N	Y
	0C: SET PLUGIN ENABLE	Enable/disable a PLUGIN	N	Y
	0D: SET PLUGIN LOCK	Lock/unlock PLUGINS	N	Y
	0E: UNMAP CONTROL	Un-map a learned control	Y	N
	0F: SET MAPPED CONTROL NAME	Set the name of a mapped control	Y	N
	10: MACRO PLUGIN PAGE INDEX	Current MACRO PLUGIN page index	N	Y
	11: CONTROL LEARNED	A param has been learned and assigned to a control	N	Y
	12: TOGGLE REMOTE PAGE	Request the DAW to toggle the remote page in a PLUGIN	N	Y
	13: SET PLUGIN CONTROL DETAILS	Set a PLUGIN control details	Y	N
	14: RESET PLUGIN CONTROL DETAILS	Reset (clear) a PLUGIN control details	Y	N
	15: SET TRACK SELECT MODE	Indicates PLUGIN track select mode has been entered	N	Y
	16: SET PLUGIN ENABLE MODE	Indicates PLUGIN enable mode has been entered	N	Y

	17: SET PLUGIN SELECT MODE	Indicates PLUGIN select mode has been entered	N	Y
	18: RFU	Reserved for future use	-	-
	19: PLUGIN PARAM SWEEP	Start or continue the PLUGIN param sweep used in param learning	Y	N
	1A: PLUGIN PARAM SWEEP VALUE	The next PLUGIN param sweep value in the learning sequence	N	Y
	1B: RFU	Reserved for future use	-	-
	1C: PLUGIN PARAM SWEEP END	The PLUGIN param has been learned, and the param sweep has ended	N	Y
	1D: PLUGIN PARAM SWEEP RESTART	Restart the PLUGIN param sweep sequence	Y	N
0C: MIX	01: SET MIX ALL TRACKS MODE	Set MIX into “all tracks” mode	N	Y
	02: SET MIX TRACK MODE	Set MIX into “single track” mode	N	Y
	03: NUM SEND TRACKS	The number of available send tracks	Y	N
	04: DAW SELECT TRACK	The DAW has selected a track	Y	N
	05: SET ALL TRACKS MODE	Set the “all tracks” mode	N	Y
	06: TOGGLE TRACK GROUPING	Toggle track grouping on the currently selected track	N	Y
	07: REQUEST SEND TRACK NAMES	Request a set of 8 send track names	N	Y
	08: SEND TRACK NAMES	A set of 8 send track names	Y	N
	09: REFRESH MIX MODE	Refresh MIX mode in ROTO	Y	N
	0A: DAW SELECT FOCUS TRACK	The DAW has selected the focus track	Y	N
	0B: SET MIX VU METER THRESHOLDS	Set the MIX VU meter threshold points	Y	N

2. GENERAL Commands: 0A

2.1 DAW STARTED: 01

Command
F0 00 22 03 02 0A 01 F7

Description
This is sent by the DAW when it first starts (or a new session is started). The DAW should not send any further commands until it receives PING DAW from ROTO.

Direction	
TO ROTO	Y
FROM ROTO	N

2.2 PING DAW: 02

Command
F0 00 22 03 02 0A 02 F7

Description
This is sent by ROTO to check if a DAW is connected – it will be sent by ROTO after receiving DAW STARTED or when ROTO is plugged into a PC. The DAW should respond with DAW PING RESPONSE when this is received. The DAW should not send any further commands until it receives ROTO-DAW CONNECTED from ROTO.
Note: ROTO will send this command every second for 1 minute to poll for a connected DAW.

Direction	
TO ROTO	N
FROM ROTO	Y

2.3 DAW PING RESPONSE: 03

Command
F0 00 22 03 02 0A 03 <DT> F7
DT = DAW Type: Ableton Live (01), Bitwig Studio (02), Logic Pro (03)

Description
This command should be sent by the DAW when it receives the PING DAW command from ROTO.

Direction	
TO ROTO	Y
FROM ROTO	N

2.4 NUM TRACKS: 04

Command
F0 00 22 03 02 0A 04 <NT:2> F7
NT = Number of tracks, formatted as two 7-bit values

Description	
-------------	--

This command is sent from the DAW to indicate the number of tracks – either the number of audio or master-return tracks depending on the mixer mode.	
--	--

Direction	
-----------	--

TO ROTO	Y
FROM ROTO	N

2.5 FIRST TRACK: 05

Command

F0 00 22 03 02 0A 05 <FT:2> F7

FT = First track index in multiples of 8 (0000 = Track page 1, 0008 = Track page 2, etc.), formatted as two 7-bit values
--

Description	
-------------	--

This command is sent from the DAW to indicate the current tracks page.
--

Direction	
-----------	--

TO ROTO	Y
FROM ROTO	N

2.6 SET FIRST TRACK: 06

Command

F0 00 22 03 02 0A 06 <FT:2> F7

FT = First track index in multiples of 8 (0000 = Track page 1, 0008 = Track page 2, etc.), formatted as two 7-bit values
--

Description	
-------------	--

This command is sent from ROTO to indicate when the page of tracks has been changed.
--

Direction	
-----------	--

TO ROTO	N
FROM ROTO	Y

2.7 TRACK DETAILS: 07

Command

F0 00 22 03 02 0A 07 <TI:2 TN:0D CS GT> F7
--

TI = Track index, formatted as two 7-bit values

TN = Track name: 0D-byte NULL terminated ASCII string, padded with 00s if needed
--

CS = Colour scheme: 00 - 52 (see Appendix A)
--

GT = Grouped track: NO (00), YES (01)

Description

This command is sent from the DAW for each track on the current tracks page.
--

Direction	
TO ROTO	Y
FROM ROTO	N

2.8 TRACK DETAILS END: 08

Command
F0 00 22 03 02 0A 08 F7

Description
This command is sent from the DAW once all TRACK DETAILS commands have been sent.

Direction	
TO ROTO	Y
FROM ROTO	N

2.9 ROTO SELECT TRACK: 09

Command
F0 00 22 03 02 0A 09 <TI:2> F7
TI = Index of the track to select, formatted as two 7-bit values

Description
This command is sent when a track is selected via ROTO.

Direction	
TO ROTO	N
FROM ROTO	Y

2.10 REQUEST TRANSPORT STATUS: 0A

Command
F0 00 22 03 02 0A 0A F7

Description
This command is sent from ROTO to request the current transport controls status.

Direction	
TO ROTO	N
FROM ROTO	Y

2.11 TRANSPORT STATUS: 0B

Command
F0 00 22 03 02 0A 0B <PS SS RS SR LS PI PO AS> F7
PS = Play status: Off (00), On (01)
SS = Stop status: Off (00), always this value
RS = Record status: Off (00), On (01)
SR = Session Record status: Off (00), On (01)

LS = Loop status: Off (00), On (01)
 PI = Punch-in status: Off (00), On (01)
 PO = Punch-out status: Off (00), On (01)
 AS = Re-enable automation status: Off (00), On (01)

Description

This command is sent from the DAW in response to the REQUEST TRANSPORT STATUS command.
 Note: No response is sent for a Logic Pro DAW, the transport status is instead sent in the form of MIDI CCs.

Direction

TO ROTO	Y
FROM ROTO	N

2.12 ROTO-DAW CONNECTED: 0C

Command

F0 00 22 03 02 0A 0C F7

Description

This is sent by ROTO once it has received and finished processing DAW PING RESPONSE from a DAW. Once the DAW has received this command it can send other commands to ROTO as needed.

Direction

TO ROTO	N
FROM ROTO	Y

2.13 REQUEST ROTO FW VERSION: 0D

Command

F0 00 22 03 02 0A 0D F7

Description

This command is sent from the DAW to request the ROTO firmware version.

Direction

TO ROTO	Y
FROM ROTO	N

2.14 ROTO FW VERSION: 0E

Command

F0 00 22 03 02 0A 0E <VX VY VZ GC:7> F7

VX = ROTO-CONTROL major version

VY = ROTO-CONTROL minor version

VZ = ROTO-CONTROL patch version

GC = Short GIT commit in ASCII bytes

Description

This command is sent from ROTO after receiving the REQUEST ROTO FW VERSION command.

Direction	
TO ROTO	N
FROM ROTO	Y

2.15 REQUEST ROTO SYSEX API VERSION: 0F

Command
F0 00 22 03 02 0A 0F F7

Description
This command is sent from the DAW to request the ROTO SYSEX API version.

Direction	
TO ROTO	Y
FROM ROTO	N

2.16 ROTO SYSEX API VERSION: 10

Command
F0 00 22 03 02 0A 10 <VX VY> F7 VX = ROTO-CONTROL SYSEX API major version VY = ROTO-CONTROL SYSEX API minor version

Description
This command is sent from ROTO after receiving the REQUEST ROTO SYSEX API VERSION command.

Direction	
TO ROTO	N
FROM ROTO	Y

2.17 SET TRACK DETAILS: 11

Command
F0 00 22 03 02 0A 11 <TI:2 TN:0D CS GT> F7 TI = Track index, formatted as two 7-bit values TN = Track name: 0D-byte NULL terminated ASCII string, padded with 00s if needed CS = Colour scheme: 00 - 52 (see Appendix A) GT = Grouped track: NO (00), YES (01)

Description
This command is sent from the DAW for each track on the current tracks page. This command is the same as the TRACK DETAILS command, except it updates the shown track on ROTO immediately.

Direction	
TO ROTO	Y

FROM ROTO	N
-----------	----------

2.18 RESET TRACK DETAILS: 12

Command
F0 00 22 03 02 0A 12 <TI:2> F7
TI = Track index, formatted as two 7-bit values

Description
This command is sent from the DAW to remove the track with the specified track index. The screen showing the track on ROTO is cleared immediately.

Direction	
TO ROTO	Y
FROM ROTO	N

2.19 SET TRACK COLOUR: 13

Command
F0 00 22 03 02 0A 13 <TI:2 CR:2 CG:2 CB:2> F7
TI = Track index, formatted as two 7-bit values
CR = The RGB red colour component, formatted as two 7-bit values
CG = The RGB green colour component, formatted as two 7-bit values
CB = The RGB blue colour component, formatted as two 7-bit values

Description
This command is sent from the DAW to set the track colour in RGB.

Direction	
TO ROTO	Y
FROM ROTO	N

2.20 ROTO PAGE LEFT: 14

Command
F0 00 22 03 02 0A 14 F7

Description
This command is sent from ROTO when the left button has been pressed. Currently only sent if ROTO is connected to a Logic Pro DAW, in MIX mode, showing track focus.

Direction	
TO ROTO	N
FROM ROTO	Y

2.21 ROTO PAGE RIGHT: 15

Command
F0 00 22 03 02 0A 15 F7

Description
This command is sent from ROTO when the right button has been pressed. Currently only sent if ROTO is connected to a Logic Pro DAW, in MIX mode, showing track focus.

Direction	
TO ROTO	N
FROM ROTO	Y

2.21 SET CURRENT TRACK NAME: 16

Command
F0 00 22 03 02 0A 16 <TN:0D> F7
TN = Track name: 0D-byte NULL terminated ASCII string, padded with 00s if needed

Description
This command is sent from the DAW to set the currently selected track name.

Direction	
TO ROTO	Y
FROM ROTO	N

2.22 SET CURRENT TRACK COLOUR: 17

Command
F0 00 22 03 02 0A 17 <CR:2 CG:2 CB:2> F7
CR = The RGB red colour component, formatted as two 7-bit values
CG = The RGB green colour component, formatted as two 7-bit values
CB = The RGB blue colour component, formatted as two 7-bit values

Description
This command is sent from the DAW to set the currently selected track colour in RGB.

Direction	
TO ROTO	Y
FROM ROTO	N

2.23: SHOW CONTROL VALUE: 18

Command
F0 00 22 03 02 0A 18 <CT CI CV:0D> F7
CT = Control type: Knob (00), Switch (01)
CI = Index of the control: 00 - 07
CV = Control value: 0D-byte NULL terminated ASCII string, padded with 00s if needed

Description
This command is sent from the DAW to show the current value of the specified control on the control's screen. ROTO uses a timeout to reset the screen back to the control name, if no further SHOW CONTROL VALUE commands are received for that control.

Direction	
TO ROTO	Y
FROM ROTO	N

3. PLUGIN Commands: 0B

3.1 SET PLUGIN MODE: 01

Command
F0 00 22 03 02 0B 01 F7

Description
This command is sent from ROTO when PLUGIN mode is selected.

Direction	
TO ROTO	N
FROM ROTO	Y

3.2 NUM PLUGINS: 02

Command
F0 00 22 03 02 0B 02 <NP> F7
NP = Number of PLUGINS on the current track

Description
This command is sent from the DAW to indicate how many PLUGINS are on the currently selected track.

Direction	
TO ROTO	Y
FROM ROTO	N

3.3 FIRST PLUGIN: 03

Command
F0 00 22 03 02 0B 03 <FP> F7
FP = First PLUGIN index in multiples of 8 (00 = PLUGIN page 1, 08 = PLUGIN page 2, etc.)

Description
This command is sent from the DAW to indicate the current PLUGIN page.

Direction	
TO ROTO	Y
FROM ROTO	N

3.4 SET FIRST PLUGIN: 04

Command
F0 00 22 03 02 0B 04 <FP> F7
FP = First PLUGIN index in multiples of 8 (00 = PLUGIN page 1, 08 = PLUGIN page 2, etc.)

Description
This command is sent from ROTO to indicate when the page of PLUGINS has been changed.

Direction	
TO ROTO	N
FROM ROTO	Y

3.5 PLUGIN DETAILS: 05

Command
F0 00 22 03 02 0B 05 <PI PH:8 PE PN:0D PT NP> F7 PI = PLUGIN index PH = PLUGIN hash PE = PLUGIN enabled status: Disabled (00), Enabled (01) PN = PLUGIN name: 0D-byte NULL terminated ASCII string, padded with 00s if needed PT = PLUGIN type: Normal (00), MACRO PLUGIN (01), Third-party PLUGIN (02) NP = Number of MACRO PLUGIN control pages, set to 00 if not a MACRO PLUGIN. Ignored if the connected DAW is Ableton Live

Description
This command is sent from the DAW for each PLUGIN on the current PLUGIN page.

Direction	
TO ROTO	Y
FROM ROTO	N

3.6 PLUGIN DETAILS END: 06

Command
F0 00 22 03 02 0B 06 F7

Description
This command is sent from the DAW once all PLUGIN DETAILS commands have been sent.

Direction	
TO ROTO	Y
FROM ROTO	N

3.7 ROTO SELECT PLUGIN: 07

Command
F0 00 22 03 02 0B 07 <PI> F7 PI = Index of the PLUGIN to select

Description
This command is sent from ROTO when a PLUGIN is selected.

Direction	
TO ROTO	N
FROM ROTO	Y

3.8 DAW SELECT PLUGIN: 08

Command
F0 00 22 03 02 0B 08 <PI MP FS> F7 PI = Index of the PLUGIN to select MP = MACRO PLUGIN page index (zero-based), set to 00 if not a MACRO PLUGIN. Ignored if the connected DAW is Ableton Live FS = Force selection of the PLUGIN, overriding PLUGIN lock if needed: No force (00), Force (01)

Description
This command is sent from the DAW when a PLUGIN is selected.

Direction	
TO ROTO	Y
FROM ROTO	N

3.9 SET DEVICE LEARN: 09

Command
F0 00 22 03 02 0B 09 <DL> F7 DL = Device learn status: Off (00), On (01)

Description
This command is sent from ROTO when entering or exiting learn mode.

Direction	
TO ROTO	N
FROM ROTO	Y

3.10 LEARN PARAM: 0A

Command
F0 00 22 03 02 0B 0A <PI:2 PH:6 MP CI NS PP:2 PN:0D SN:NS*0D> F7 PI = Param index PH = Param hash MP = MACRO param: NO (00), YES (01) CI = Add a centre (top) indent: NO (00), YES (01). If this is set to YES, the number of steps (NS) setting is ignored. Only applies to knob controls. NS = Number of steps: 00 or 02 - 18 PP = Param position: 0 - 16383, formatted as two 7-bit values PN = Param name: 0D-byte NULL terminated ASCII string, padded with 00s if needed SN = If NS is <= 10: An array of NS x 0D-byte NULL terminated ASCII strings, each string padded with 00s if needed If NS > 10: No data

Description
This command is sent from the DAW when in learn mode and a PLUGIN param has been selected in the DAW to be learned. It is also sent from the DAW in response to each CONTROL MAPPED command received via ROTO.

Direction	
TO ROTO	Y
FROM ROTO	N

3.11 CONTROL MAPPED: 0B

Command
F0 00 22 03 02 0B 0B <PI:2 PH:6 CT CI MP> F7 PI = Param index PH = Param hash CT = Control type: Knob (00), Switch (01) CI = Index of the control within the current PLUGIN page MP = MACRO param: NO (00), YES (01)

Description
This command is sent from ROTO for each control that is mapped on the current PLUGIN page.

Direction	
TO ROTO	N
FROM ROTO	Y

3.12 SET PLUGIN ENABLED: 0C

Command
F0 00 22 03 02 0B 0C <PI ES> F7 PI = PLUGIN index ES = PLUGIN enabled status: Disabled (00), Enabled (01)

Description
This command is sent from ROTO when the selected PLUGIN is enabled or disabled.

Direction	
TO ROTO	N
FROM ROTO	Y

3.13 SET PLUGINS LOCK: 0D

Command
F0 00 22 03 02 0B 0D <LS> F7 LS = PLUGINS lock status: Unlocked (00), Locked (01)

Description
This command is sent from ROTO when the selected PLUGIN is locked or unlocked.

Direction	
TO ROTO	N
FROM ROTO	Y

3.14 UNMAP CONTROL: 0E

Command
F0 00 22 03 02 0B 0E <CT CI> F7 CT = Control type: Knob (00), Switch (01) CI = Index of the control within the current PLUGIN page

Description
This command can be sent from a DAW to unmap a mapped control on the current controls page.

Direction	
TO ROTO	Y
FROM ROTO	N

3.15 SET MAPPED CONTROL NAME: 0F

Command
F0 00 22 03 02 0B 0F <PI:2 PH:6 PN:0D> F7 PI = Param index PH = Param hash PN = Param name: 0D-byte NULL terminated ASCII string, padded with 00s if needed

Description
This command can be sent from a DAW to update the displayed name of a macro param.

Direction	
TO ROTO	Y
FROM ROTO	N

3.16 MACRO PLUGIN PAGE INDEX: 10

Command
F0 00 22 03 02 0B 10 <PI> F7 PI = MACRO PLUGIN Page index (zero based). Not sent if the plugin is not a MACRO PLUGIN.

Description
This command is sent from ROTO when a MACRO PLUGIN is selected, and the page of controls is changed via the arrow keys

Direction	
TO ROTO	N
FROM ROTO	Y

3.17 CONTROL LEARNED: 11

Command
F0 00 22 03 02 0B 11 <CT CI> F7 CT = Control type: Knob (00), Switch (01) CI = PLUGIN control index (00 - 3F)

Description	
This command is sent from ROTO when a control (knob or switch) has been successfully learned.	

Direction	
TO ROTO	N
FROM ROTO	Y

3.18 TOGGLE REMOTE PAGE: 12

Command
F0 00 22 03 02 0B 12 F7

Description
This command is sent from ROTO to request the DAW to toggle the remote page in a PLUGIN.

Direction	
TO ROTO	N
FROM ROTO	Y

3.19 SET PLUGIN CONTROL DETAILS: 13

Command
F0 00 22 03 02 0B 13 <CT CI CN:0D CS> > F7 CT = Control type: Knob (00), Switch (01) CI = PLUGIN control index (00 - 3F) CN = PLUGIN control name: 0D-byte NULL terminated ASCII string, padded with 00s if needed CS = Colour scheme: 00 - 52 (see Appendix A)

Description
This command is sent to ROTO to show the details for a PLUGIN control, used normally in Logic Pro SMART mode.

Direction	
TO ROTO	Y
FROM ROTO	N

3.20 RESET PLUGIN CONTROL DETAILS: 14

Command
F0 00 22 03 02 0B 14 <CT CI> > F7 CT = Control type: Knob (00), Switch (01) CI = PLUGIN control index (00 - 3F)

Description
This command is sent to ROTO to reset (clear) the details for a PLUGIN control, used normally in Logic Pro SMART mode.

Direction

TO ROTO	Y
FROM ROTO	N

3.21 SET TRACK SELECT MODE: 15

Command
F0 00 22 03 02 0B 15 F7

Description
This command is sent from ROTO to indicate track select (FUNC) has been entered from PLUGIN mode. Note: Logic Pro only.

Direction	
TO ROTO	N
FROM ROTO	Y

3.22 SET PLUGIN ENABLE MODE: 16

Command
F0 00 22 03 02 0B 16 F7

Description
This command is sent from ROTO to indicate PLUGIN enable has been entered. Note: Logic Pro only.

Direction	
TO ROTO	N
FROM ROTO	Y

3.23 SET PLUGIN SELECT MODE: 17

Command
F0 00 22 03 02 0B 17 F7

Description
This command is sent from ROTO to indicate PLUGIN select mode has been entered. Note: Logic Pro only.

Direction	
TO ROTO	N
FROM ROTO	Y

3.24 RFU: 18

This command is unused and reserved for future use.

3.25 PLUGIN PARAM SWEEP: 19

Command

F0 00 22 03 02 0B 19 <PI:2> > F7
PP = Param index, formatted as two 7-bit values

Description

This command is sent to ROTO to start and continue the param sweep of a param to learn. Note: Logic Pro only.

Direction

TO ROTO	Y
FROM ROTO	N

3.26 PLUGIN PARAM SWEEP VALUE: 20

Command

F0 00 22 03 02 0B 20 <MC CP:2 CV:2> > F7
MC = The MIDI channel of the PLUGIN param being learned
CP = The CC param (14-bit) of the PLUGIN param being learned, LSB followed by the MSB
CV = The CC sweep value (14-bit) of the PLUGIN param being learned, LSB followed by the MSB

Description

This command is from ROTO in response to the PLUGIN PARAM SWEEP command. The sweep value sent follows a specific pattern of values so that a PLUGIN param can be learned by ROTO. Note: Logic Pro only.

Direction

TO ROTO	N
FROM ROTO	Y

3.27 RFU: 21

This command is unused and reserved for future use.

3.28 PLUGIN PARAM SWEEP END: 22

Command

F0 00 22 03 02 0B 22 > F7

Description

This command is from ROTO once the PLUGIN param has been successfully learned, indicating the end of the param sweep process. Note: Logic Pro only.

Direction

TO ROTO	N
FROM ROTO	Y

3.29 PLUGIN PARAM SWEEP RESTART: 23

Command

F0 00 22 03 02 0B 23 <PI:2> > F7

PP = Param index, formatted as two 7-bit values

Description

This command is sent to ROTO to abort any current PLUGIN param sweep process, and to restart the sweep sequence. Note: Logic Pro only.

Direction

TO ROTO

Y

FROM ROTO

N

4. MIX Commands: 0C

4.1 SET MIX ALL TRACKS MODE: 01

Command
F0 00 22 03 02 0C 01 <AM KM SM SI> F7 AM = All tracks mode: Audio (00), Master-Return (01) KM = Knob mode: Level (00), Pan (01), Send (02) SM = Switch mode: Mute (00), Solo (01), Arm Recording (02) SI = Send index: The (zero-based) index of the send to control if the Knob mode is Send. Ranges from 0 - Number of available sends

Description
This command is sent from ROTO to set the mixer in MIX all tracks mode.

Direction	
TO ROTO	N
FROM ROTO	Y

4.2 SET MIX TRACK MODE: 02

Command
F0 00 22 03 02 0C 02 <CI> F7 CI = Track control page index in multiples of 8 (00 = Track controls page 1, 08 = Track controls page 2, etc.)

Description
This command is sent from ROTO to set the mixer in MIX track mode.

Direction	
TO ROTO	N
FROM ROTO	Y

4.3 NUM SEND TRACKS: 03

Command
F0 00 22 03 02 0C 03 <NS> F7 NS = Number of available send tracks

Description
This command is sent from the DAW to indicate how many send tracks are available.

Direction	
TO ROTO	Y
FROM ROTO	N

4.4 DAW SELECT TRACK: 04

Command
F0 00 22 03 02 0C 04 <TI:2 TN:0D CS GT> F7

TI = Track index, formatted as two 7-bit values
 TN = Track name: 0D-byte NULL terminated ASCII string, padded with 00s if needed
 CI = Colour scheme: 00 - 52 (see Appendix A)
 GT = Grouped track: NO (00), YES (01)

Description

This command is sent from the DAW when a mixer track is selected.

Direction

TO ROTO	Y
FROM ROTO	N

4.5 SET ALL TRACKS MODE: 05

Command

F0 00 22 03 02 0C 05 <AM> F7
 AM = All tracks mode: Audio (00), Master-Return (01)

Description

This command is sent from ROTO when the MIX all tracks mode is changed to show audio or master-return tracks.

Direction

TO ROTO	N
FROM ROTO	Y

4.6 TOGGLE TRACK GROUPING: 06

Command

F0 00 22 03 02 0C 06 <TI:2> F7
 TI = Track index formatted as two 7-bit values

Description

This command is sent from ROTO to toggle track grouping on the currently selected track (if grouping is applicable for that track).

Direction

TO ROTO	N
FROM ROTO	Y

4.7 REQUEST SEND TRACK NAMES: 07

Command

F0 00 22 03 02 0C 07 <SI> F7
 SI = SEND names index (zero based)

Description

Request a block of 8 send track names from the DAW containing the send track names index.

Direction	
TO ROTO	N
FROM ROTO	Y

4.8 SEND TRACK NAMES: 08

Command
F0 00 22 03 02 0C 08 <SI SN:08*0D> > F7 SI = SEND names index (zero based) SN = An array of 8 x 0D-byte NULL terminated ASCII strings, each string padded with 00s if needed

Description
This command should be sent by the DAW after receiving either the SET MIX ALL TRACKS MODE, SET MIX TRACK MODE, or REQUEST SEND TRACK NAMES commands and at least one send track is present in the DAW: - SET MIX ALL TRACKS MODE command is received: It must send a block of 8 send track names with the start index being a multiple of 8 and containing the send index received in the command. Examples: - Send index is 00, the send track names sent should be from index 00. - Send index is 07, the send track names sent should be from index 00. - Send index is 08, the send track names sent should be from index 08. - Send index is 14, the send track names sent should be from index 10. Note: If the track with the send index is renamed, the block of 8 track names must be re-sent to ROTO. - SET MIX TRACK MODE command is received: It must send a block of 8 SEND names with the start index being calculated based on the track control page index received in the command. Examples: - Track control page index is 00, the send track names sent should be from index 00. - Track control page index is 08, the send track names sent should be from index 06. - Track control page index is 10, the send track names sent should be from index 0E. - REQUEST SEND TRACK NAMES command is received: It must send a block of 8 send track names with the start index being a multiple of 8 and containing the send index received in the command. Note: If there is no send track name then set all 00s for that string.

Direction	
TO ROTO	Y
FROM ROTO	N

4.9 REFRESH MIX MODE: 09

Command
F0 00 22 03 02 0C 09 F7

Description

This command is sent from the DAW to refresh MIX mode in ROTO (if ROTO is in MIX mode).

Direction	
TO ROTO	Y
FROM ROTO	N

4.10 DAW SELECT FOCUS TRACK: 0A

Command
F0 00 22 03 02 0C 0A <TI:2 CR:2 CG:2 CB:2> F7 TI = Track index, formatted as two 7-bit values CR = The RGB red colour component, formatted as two 7-bit values CG = The RGB green colour component, formatted as two 7-bit values CB = The RGB blue colour component, formatted as two 7-bit values

Description
This command is sent from the DAW to select the focus track.

Direction	
TO ROTO	Y
FROM ROTO	N

4.11 SET MIX VU METER THRESHOLDS: 0B

Command
F0 00 22 03 02 0C 0B <YT RT> F7 YT = The green to yellow threshold: 00 - 7F RT = The yellow to red threshold: 00 - 7F

Description
This command is sent from the DAW to set the yellow and red threshold points in the MIX VU meters. Note that the red threshold must always be larger than the yellow threshold. Note: This command applies to all MIX VU meters.

Direction	
TO ROTO	Y
FROM ROTO	N

Appendix A: Colour Scheme Table

The Colour Scheme table is specified below, with the background colour in RGB and the text colour as black or white.

Index (decimal)	Background Colour			Text Colour
	Red	Green	Blue	
0	FF	94	A0	Black
1	FF	A5	27	Black
2	D3	99	25	Black
3	FF	F4	77	Black
4	C6	FB	00	Black
5	1B	FF	2D	Black
6	26	FF	A2	Black
7	5F	FF	DF	Black
8	90	C5	F6	Black
9	57	80	DB	Black
10	97	A7	F6	Black
11	E0	6C	DB	Black
12	ED	53	9A	Black
13	FF	FF	F6	Black
14	FF	36	34	Black
15	FF	6C	02	Black
16	9E	62	48	Black
17	FF	F0	32	Black
18	8C	FF	63	Black
19	3F	C3	00	Black
20	00	BF	A8	Black
21	19	E9	F6	Black
22	10	A4	E5	Black
23	00	7D	B9	Black
24	8D	6C	DB	Black
25	BD	77	BF	Black
26	FF	39	CC	Black
27	D8	D0	C8	Black
28	EC	68	56	Black
29	FF	A3	6F	Black
30	DB	AD	6D	Black
31	F6	FF	A7	Black
32	DA	E4	92	Black
33	C1	D0	6F	Black
34	A1	C4	88	Black
35	DC	FD	D9	Black
36	D4	F1	EF	Black
37	C0	C1	DA	Black
38	D4	BB	DB	Black

39	B4	98	DC	Black
40	ED	DC	D9	Black
41	AF	A9	A3	Black
42	EE	92	86	Black
43	BE	82	52	Black
44	9E	83	66	Black
45	C6	BA	65	Black
46	AC	BE	00	Black
47	81	BO	4A	Black
48	8D	C2	B3	Black
49	A1	B3	BD	Black
50	8A	A5	BF	Black
51	88	93	C4	Black
52	AB	95	AE	Black
53	C6	9F	B7	Black
54	C3	71	90	Black
55	7F	7B	76	Black
56	B5	33	31	White
57	AF	51	2F	White
58	76	4F	3E	White
59	E3	C3	00	White
60	8A	96	1D	White
61	56	9F	2F	White
62	0A	9C	88	White
63	24	63	7F	White
64	1B	2F	90	White
65	30	52	9C	White
66	65	4B	A6	White
67	A9	4B	A6	White
68	D3	2E	6A	White
69	3E	3C	39	White
70	10	10	10	White
71	FF	00	00	White
72	00	FF	00	Black
73	FF	FF	00	Black
74	00	00	F6	White
75	FF	00	F6	White
76	00	FF	F6	Black
77	84	00	00	White
78	84	80	00	Black
79	00	80	00	White
80	00	80	7B	Black
81	00	00	7B	White
82	84	00	7B	White

Appendix B: MIDI CC Mappings

B.1 MIX MODE MIDI CC Mappings

The table below defines the MIDI CC mappings in MIX mode.

Note 1: All MIDI CC messages use MIDI channel 16.

Note 2: All knobs return 14-bit MIDI values, and switches 7-bit MIDI values.

Control	MIDI CC Param (decimal)	Notes
MIX knob 1	12 + 44	
MIX knob 2	13 + 45	
MIX knob 3	14 + 46	
MIX knob 4	15 + 47	
MIX knob 5	16 + 48	
MIX knob 6	17 + 49	
MIX knob 7	18 + 50	
MIX knob 8	19 + 51	
MIX switch 1	20	
MIX switch 2	21	
MIX switch 3	22	
MIX switch 4	23	
MIX switch 5	24	
MIX switch 6	25	
MIX switch 7	26	
MIX switch 8	27	
TRANSPORT switch 1	28	
TRANSPORT switch 2	29	
TRANSPORT switch 3	30	
TRANSPORT switch 4	31	
TRANSPORT switch 5	32	
TRANSPORT switch 6	33	
TRANSPORT switch 7	34	
TRANSPORT switch 8	35	
MIX knob 1 Touch	52	Touch ON (7F), Touch OFF (00)
MIX knob 2 Touch	53	"
MIX knob 3 Touch	54	"
MIX knob 4 Touch	55	"
MIX knob 5 Touch	56	"
MIX knob 6 Touch	57	"
MIX knob 7 Touch	58	"
MIX knob 8 Touch	59	"
MIX track level at 0dB	64	
MIX track 1 VU Meter	65 + 66	Left and Right meter values
MIX track 2 VU Meter	67 + 68	"
MIX track 3 VU Meter	69 + 70	"
MIX track 4 VU Meter	71 + 72	"
MIX track 5 VU Meter	73 + 74	"

MIX track 6 VU Meter	75 + 76	“
MIX track 7 VU Meter	77 + 78	“
MIX track 8 VU Meter	79 + 80	“
Logic Pro RESERVED	81 to 127	Logic Pro DAW only

B.2 PLUGIN MODE MIDI CC Mappings

ALL DAWS EXCEPT LOGIC PRO

The table below defines the MIDI CC mappings in PLUGIN mode.

Note 1: All MIDI CC messages use MIDI channel 16.

Note 2: All knobs return 14-bit MIDI values, and switches 7-bit MIDI values.

Control	MIDI CC Param (decimal)	Notes
PLUGIN knob 1	12 + 44	
PLUGIN knob 2	13 + 45	
PLUGIN knob 3	14 + 46	
PLUGIN knob 4	15 + 47	
PLUGIN knob 5	16 + 48	
PLUGIN knob 6	17 + 49	
PLUGIN knob 7	18 + 50	
PLUGIN knob 8	19 + 51	
PLUGIN switch 1	20	
PLUGIN switch 2	21	
PLUGIN switch 3	22	
PLUGIN switch 4	23	
PLUGIN switch 5	24	
PLUGIN switch 6	25	
PLUGIN switch 7	26	
PLUGIN switch 8	27	
TRANSPORT switch 1	28	
TRANSPORT switch 2	29	
TRANSPORT switch 3	30	
TRANSPORT switch 4	31	
TRANSPORT switch 5	32	
TRANSPORT switch 6	33	
TRANSPORT switch 7	34	
TRANSPORT switch 8	35	
PLUGIN knob 1 Touch	52	Touch ON (7F), Touch OFF (00)
PLUGIN knob 2 Touch	53	“
PLUGIN knob 3 Touch	54	“
PLUGIN knob 4 Touch	55	“
PLUGIN knob 5 Touch	56	“
PLUGIN knob 6 Touch	57	“
PLUGIN knob 7 Touch	58	“
PLUGIN knob 8 Touch	59	“

LOGIC PRO DAW

Logic Pro is a little different in that for mapped parameters the MIDI CC channel and param are calculated from the learned parameter index supplied by Logic Pro. The formula is:

MIDI CC channel = 15 - (param index / 32)

MIDI CC param = 0 + (param index / 32)

This means that there are effectively 32 parameters per MIDI CC channel. All knobs and switches mapped to parameters return 14-bit MIDI values.

The table below defines the rest of the MIDI CC mappings in PLUGIN mode.

Note 1: All MIDI CC messages in this table use MIDI channel 16.

Note 2: All switches in this table return 7-bit MIDI values.

Control	MIDI CC Param (decimal)	Notes
ENABLE PLUGIN switch 1 (page 1)	20	
ENABLE PLUGIN switch 2 (page 1)	21	
ENABLE PLUGIN switch 3 (page 1)	22	
ENABLE PLUGIN switch 4 (page 1)	23	
ENABLE PLUGIN switch 5 (page 1)	24	
ENABLE PLUGIN switch 6 (page 1)	25	
ENABLE PLUGIN switch 7 (page 1)	26	
ENABLE PLUGIN switch 8 (page 1)	27	
TRANSPORT switch 1	28	
TRANSPORT switch 2	29	
TRANSPORT switch 3	30	
TRANSPORT switch 4	31	
TRANSPORT switch 5	32	
TRANSPORT switch 6	33	
TRANSPORT switch 7	34	
TRANSPORT switch 8	35	
PLUGIN knob 1 Touch	52	Touch ON (7F), Touch OFF (00)
PLUGIN knob 2 Touch	53	"
PLUGIN knob 3 Touch	54	"
PLUGIN knob 4 Touch	55	"
PLUGIN knob 5 Touch	56	"
PLUGIN knob 6 Touch	57	"
PLUGIN knob 7 Touch	58	"
PLUGIN knob 8 Touch	59	"
LEFT switch pressed	60	
RIGHT switch pressed	61	
ENABLE PLUGIN switch 1 (page 2)	62	
ENABLE PLUGIN switch 2 (page 2)	63	
ENABLE PLUGIN switch 3 (page 2)	64	
ENABLE PLUGIN switch 4 (page 2)	65	
ENABLE PLUGIN switch 5 (page 2)	66	
ENABLE PLUGIN switch 6 (page 2)	67	

ENABLE PLUGIN switch 7 (page 2)	68	
ENABLE PLUGIN switch 8 (page 2)	69	
Logic Pro RESERVED	81 to 127	